

Texas Tech field experiments for wind loads part II: meteorological instrumentation and terrain parameters

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Abstract

This paper presents information on the wind and terrain parameters at the Texas Tech Wind Engineering Research Field Laboratory. Details of the meteorological tower and instrumentation are given, as well as a description of the meteorological data which has been collected so far. The data and terrain parameters will assist researchers in performing wind tunnel simulations.

1. INTRODUCTION

Texas Tech University has established a Wind Engineering Research Field Laboratory in Lubbock, Texas. The 30 x 45 x 13 ft (9.1 x 13.7 x 4.0 m) test building and pressure measuring and data acquisition systems are described in Part I of this paper [1]. The site characteristics, wind data and terrain parameters are presented in this paper. This information will be useful to researchers in simulating air flow in wind tunnel experiments.

2. TERRAIN AND SITE CHARACTERISTICS

The field laboratory is located on Texas Tech University property in Lubbock Texas (see Fig. 1). The city of Lubbock (population 190,000) is located on the high plains of Texas (elevation 3300 ft [1000 m] above sea level, coordinates Latitude 33° N and Longitude 102° W). The plains extend for many miles in all directions, most of the land being used for agricultural production. The principal crops are cotton, grain sorghum, and wheat. The landscape is open with few trees.

The terrain surrounding the field site is flat and open. Figure 2 shows a land use map for the area near the field laboratory. The immediately surround-



Figure 1. Field site and surrounding terrain.

ing land is cultivated, the chief crop being cotton. Photographs of the terrain surrounding the field site (taken from the tower at a height of 85 ft [26 m]) are shown in Figures 3-8. The land to the north is open fields, with widely scattered small trees (less than 10 ft [3 m] high), extending several kilometers (see Figs. 3-4). Cotton fields lie immediately to the west and southwest, with suburban areas of one- and two-story houses beyond (approximately 3,000 ft [900 m] away, see Figs. 5-6). Residential areas to the south are about 5000 ft (1600 m) away. Several small research buildings are located 250 ft (75 m) to the south (see Fig. 7). A small, shallow playa lake located 800 ft (24 m) to the south-southwest can also be seen in Fig. 7. A large hospital is located 1,500 ft east-southeast of the test building (see Fig. 8), and the main university campus is roughly 1 mile (1.6 km) away in the same direction.

3. WIND CLIMATE

The field site experiences strong winds frequently throughout the year. The passage of fronts from the north and northwest in late autumn and spring brings sustained winds of 20-35 mph (9-16 m/s). Prevailing winds during the summer months are from the south to southwest at speeds of 10-20 mph (4.5-9 m/s).

Thunderstorms, common in late spring and summer, often create short duration winds of even greater magnitude. These thunderstorms can also spawn tornadoes. Lubbock is on the southwestern edge of the region known as "Tornado Alley," which extends from the Texas panhandle up through the midwest. Nearly one quarter of Lubbock was destroyed by a large tornado in May 1970. Smaller magnitude tornadoes are almost a regular occurrence in this area. So far this year, 2 tornadoes have been spotted within 10 miles (16 km) of the city of Lubbock. In 1987, one large storm spawned two tornadoes which touched down within 5 miles (8 km) of the test site.

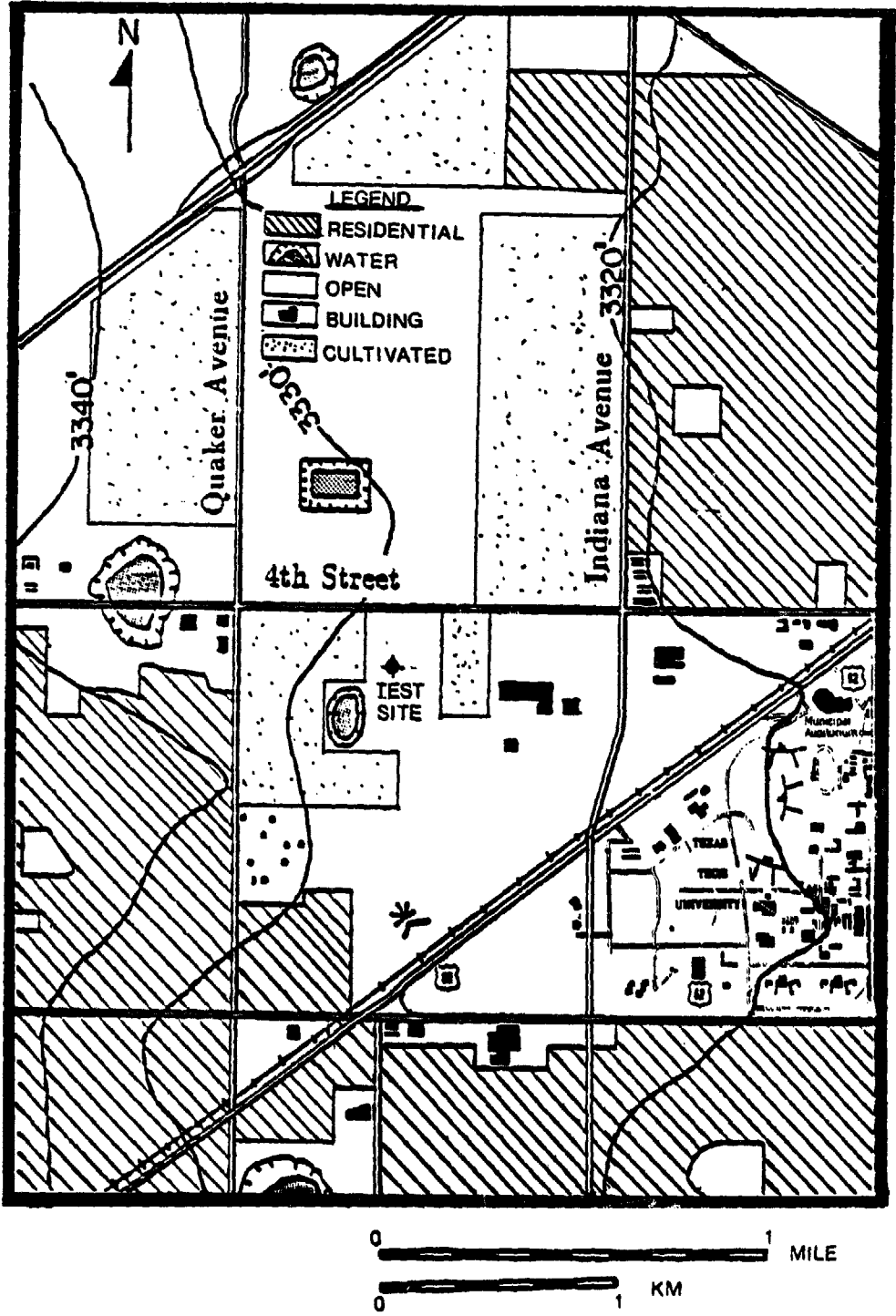


Figure 2. Map of field site and surrounding terrain.



Figure 3. Surrounding terrain looking northeast.



Figure 4. Surrounding terrain looking northwest.

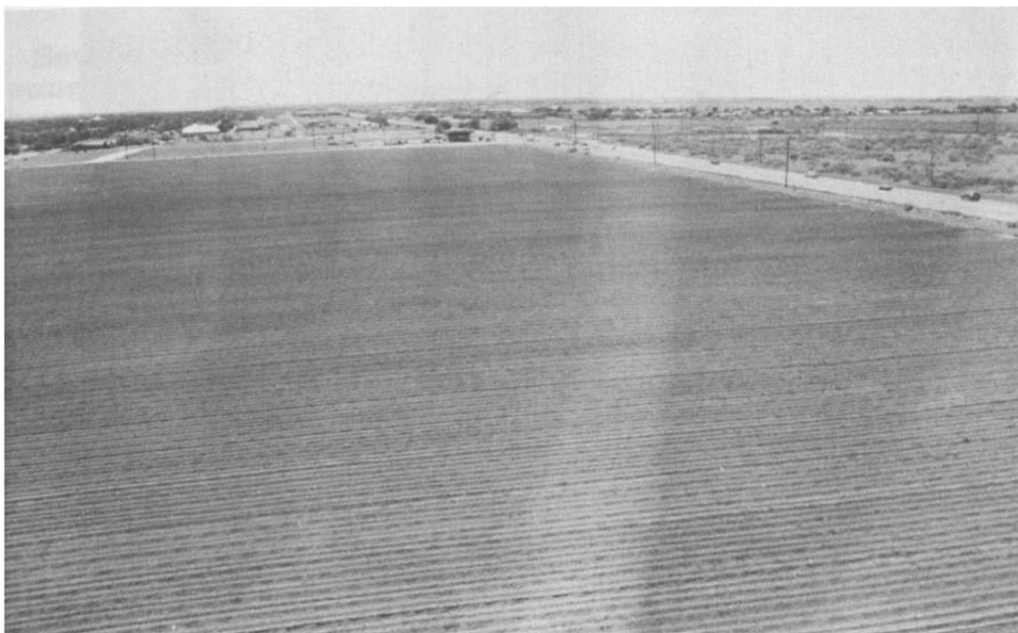


Figure 5. Surrounding terrain looking west.



Figure 6. Surrounding terrain looking southwest.



Figure 7. Surrounding terrain looking south.



Figure 8. Surrounding terrain looking east; test building is in the bottom of the photograph.

4. METEOROLOGICAL INSTRUMENTATION AND TOWER

Several different types of anemometry are used, all provided by the R. M. Young Company. Five Gill 3-cup anemometers (Model 12102) are in use. These anemometers produce an analog output voltage proportional to the wind speed. The 3-cup anemometers have a rated maximum range of 112 mph (50 m/s) and a distance constant (63% recovery) of 8.9 ft (2.7 m). Two Gill Micro-vanes (Model 12304) are used to measure wind direction. Rated delay distance (50% recovery) is 3.6 ft (1.1 m) for these wind direction sensors. Additionally, two three-component anemometers are also installed at the site (Gill UVW Anemometer Model 27005). With the optional carbon fiber thermoplastic propellers (Model 08254), these anemometers have a rated range of 90 mph (40 m/s) and a distance constant (63% recovery) of 6.9 ft (2.1 m).

Additional meteorological instrumentation is provided by Teledyne Geotech. A Model BP-100 sensor is used to record barometric pressure. Rated resolution of this sensor is 0.01 in Hg (± 0.34 hPa). Relative humidity is measured with a Model RH-200 sensor (rated accuracy $\pm 3\%$). The RH-200 sensor has a platinum temperature sensor built into the unit, which is rated $\pm 0.2^\circ\text{F}$ ($\pm 0.1^\circ\text{C}$). One additional temperature sensor (Model T-200) with similar accuracy is also available.

Meteorological instrumentation is mounted on a 160-ft (49 m) guyed tower. The tower is located at the field site, 150 ft (46 m) west (280° azimuth) of the center of the test building. Instrumentation is located at six levels on the tower: 3, 8, 13, 33, 70, and 160 ft (1, 2.5, 4, 10, 21, and 49 m). The 3-cup anemometer and direction vane at 160 ft are mounted on top of the tower. In order to minimize tower interference of wind measurements, anemometers at other levels are mounted on 6-ft (1.8 m) booms. The booms are oriented to the west northwest (300°), meaning that winds from the north, west, and south have a clean approach to the instruments. Tower interference is expected for winds from azimuth range 80° - 160° [2]. This does not pose significant problems however, since strong winds from the east and southeast are uncommon in the Lubbock area. Also, most of the nearby obstacles and buildings (including the test building) are situated in that azimuth range. Tower guy wires, which support the tower at 70 ft (21 m) (below the instrument boom) and 130 ft (40 m), are far enough away from the instruments to have little or no effect on the wind measurements.

Current anemometer placement on the tower is as follows: 3-cup anemometers are mounted at the 3, 13, 70 and 160 ft levels, wind direction vanes are placed at 13 and 160 ft, UVW anemometers are mounted at 8 and 33 ft heights. One additional 3-cup anemometer is installed atop a 13-ft (4 m) pole located halfway between the tower and the test building. This anemometer provides desired redundancy of wind speed measurement at the building roof height. Temperature, barometric pressure and relative humidity sensors are mounted at 13 ft (4 m) on the tower. A temperature sensor is also mounted at the top of the tower.

5. WIND DATA CHARACTERISTICS AND TERRAIN PARAMETERS

Several hundred data records have been collected over the past three years. The three data acquisition modes which consist of building surface pressure and meteorological data are described in Part I of this paper [1]. There is one additional data acquisition mode consisting of purely meteorological data in which a significant number of records were collected. Mode A consists of data from 3-cup anemometers at 13, 33, 70 and 160 ft (4, 10, 21 and 49 m) levels on the tower; direction vanes at 33 and 160 ft; and temperature, barometric pressure, and relative humidity at the 13-ft level. The duration of records in Mode A is also 15 minutes, with all anemometry being sampled at 10 Hz and other instruments sampled at one Hz. Sixty-three records were collected in this mode between February and April 1988, 31 of which have stationary wind speeds and wind directions at 33 ft (10 m).

The data records typically have 15-minute mean wind speeds (at building roof height of 13 ft [4 m]) of 15-25 mph (6.7-11.2 m/s). The basic wind parameters such as roughness length, power law coefficient, and turbulence intensity are computed for all records. Stationarity of the wind speed and of the wind direction is also checked for all records (see Part I [1] for procedure used). Roughly one-third to one-half of all records have stationary wind speed and wind direction. Gust spectra and integral scale are checked for selected records of interest. The wind and terrain parameters given in the rest of this section are computed from stationary records with mean wind direction from any azimuth except 80-160°.

Wind profile parameters for data collected in Mode A (1988) and Mode M04 (1989) are given in Table 1. Typical values for roughness length are about 0.6 in. (1.5 cm), in agreement with expected values for flat, open fields. A significant amount of variability is noticed in the roughness length between records. These data sets include data from different times of the year and different wind azimuths (80-160° azimuths data are not included). Winds from the north and northwest are likely equilibrium profiles, while winds from the south, southwest and northeast may still be in transition from suburban to open terrain profiles (see Fig. 2).

Turbulence intensity values are given in Table 2 for several different heights. Average longitudinal turbulence intensity at roof height is 20% for records with 15-minute mean wind speeds near 20 mph (9 m/s). It decreases with increasing height, as expected. Lateral turbulence intensity values appear to be large. These values were computed from 3-cup anemometers and direction vanes. Overshoot of the direction vanes may be yielding high lateral turbulence intensity values. A limited amount of data using a UVW anemometer taken in Mode M15 is available. These data are currently being analyzed and studied.

Longitudinal integral scale values were computed by taking the product of the mean wind speed and the time scale (integral of the wind speed autocorrelation function, see [2] for details). Values for longitudinal integral scale at different heights are given in Table 3. Length scales fall into expected ranges for open terrain.

Table 1
Wind Profile Parameters

	Mean Wind ^a Speed, mph (m/s)	Power Law Coefficient, α	Roughness Length z_0 , in. (cm)	Shear Velocity u_* , mph (m/s)
Mode A (31-record ensemble)				
Average	23.9 (10.7)	0.14	0.6 (1.5)	1.6 (0.7)
Range	18.1-29.8 (8.1-13.1)	0.10-0.17	0.02-1.5 (0.06-3.8)	1.2-2.1 (0.5-0.9)
Mode M04 (63-record ensemble)				
Average	19.8 (8.9)	0.16	0.7 (1.8)	1.4 (0.6)
Range	14.4-28.4 (6.5-12.7)	0.11-0.19	0.04-1.7 (0.09-4.4)	0.7-2.4 (0.3-1.1)

^a15-minute mean measured at 13 ft (4 m) height

Table 2
Turbulence Intensity Values

	Longitudinal Turbulence Intensities, % (Lateral Turbulence Intensities in Parentheses) ^a			
Height	13 ft (4 m)	33 ft (10 m)	70 ft (21 m)	160 ft (49 m)
Mode A (31-record ensemble)				
Average	19	17 (15)	15	14 (12)
Range	17-22	15-20 (12-18)	12-19	11-20 (8-17)
Mode M04 (69-record ensemble)				
Average	20 (19)	--	17	16
Range	15-24 (13-27)		11-22	9-22

Note: Mean wind speeds for ensembles given in Table 1.

^aLateral turbulence intensities, computed using data from 3-cup anemometers and wind direction vanes, may be overestimated due to vane overshoot.

Table 3
Longitudinal Integral Scales

Longitudinal Integral Scale, ft (m)				
Height	13 ft (4 m)	33 ft (10 m)	70 ft (21 m)	160 ft (49 m)
Mode A (31-record ensemble)				
Average	350 (107)	475 (145)	625 (191)	875 (267)
Range	130-660 (40-201)	180-840 (55-256)	280-1020 (85-311)	270-1480 (82-451)

Note: Mean wind speeds for ensemble given in Table 1.

Gust spectra have been analyzed for selected records. Figure 9 shows ensemble-averaged longitudinal gust spectra for the wind speeds at the building roof height of 13 ft (4 m) and at 33 ft (10 m). Ensembles used are the 31 stationary records of Mode A. Empirically estimated spectra [3] are also shown for reference.

6. SUMMARY

Researchers at Texas Tech University have constructed a permanent test facility to study wind effects on structures in the field. This facility is significant in that it provides a combination of capabilities not available in past field experiments. The ability to rotate the test building allows positive control over wind angle of attack. A well instrumented 160 ft (49 m) tall meteorological tower provides detailed information on the local wind regime. An advanced microcomputer-based data acquisition system allows for concurrent measurement of the more than 70 wind and pressure sensors currently available, with the possibility of further expansion.

Although the primary research emphasis currently centers on measurement and analysis of building surface pressures, many other types of experiments can be and are being performed at the field laboratory. Studies of internal pressure, flow visualization, roofing system performance, area averaging of wind loads, and structural response as well as ventilation and building exhaust are all underway at this time. A project to study vibration characteristics of traffic signal support structures is also planned at the field site.

Finally, the fact that this is a permanent facility with ongoing research distinguishes it from most of the past field experiments for wind loads on low rise structures. The permanency allows for continued improvement and refinement of measurement techniques and continuously expanding fields of study. Furthermore, it provides an opportunity for other investigators to have

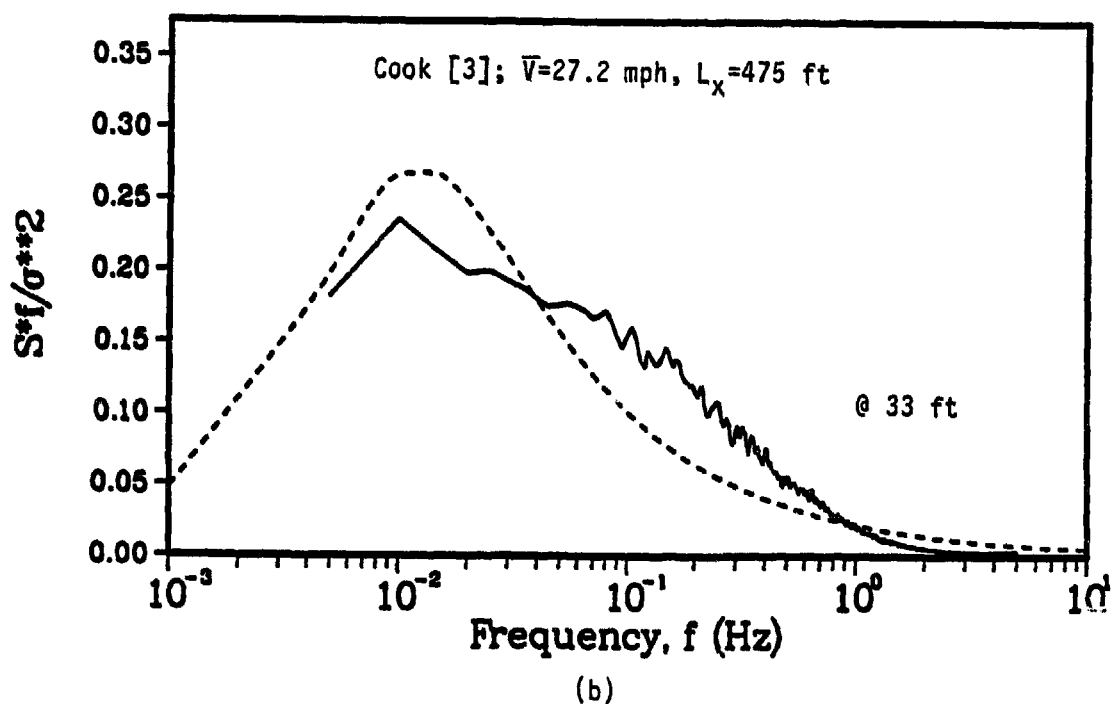
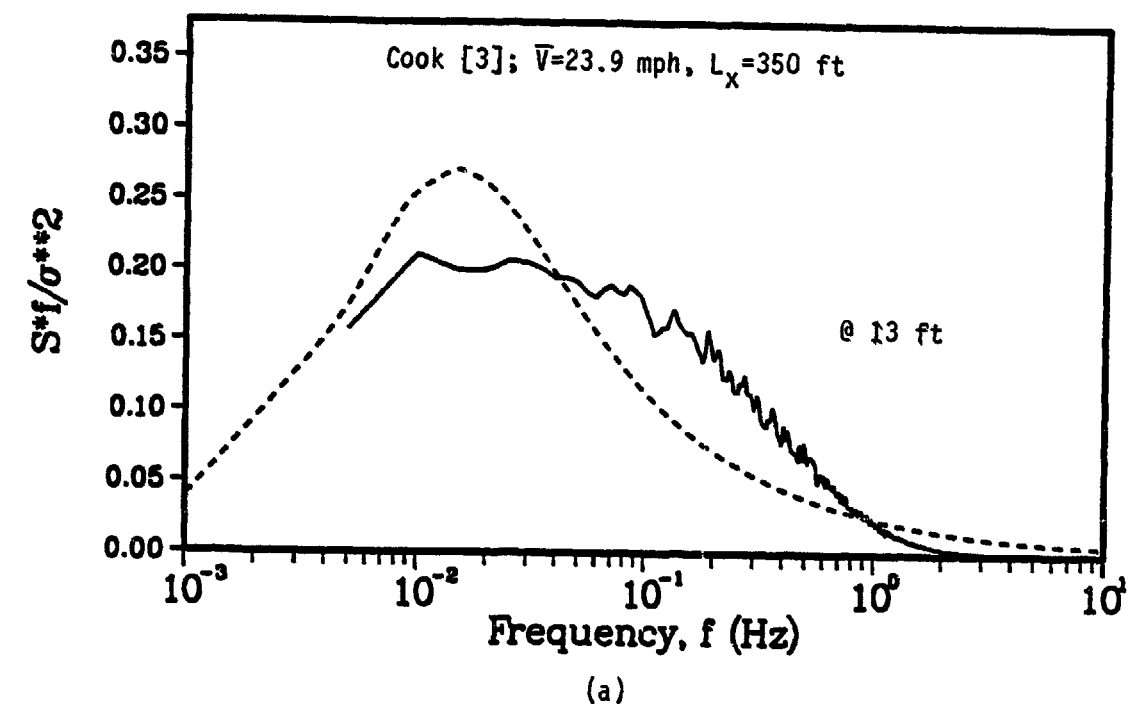


Figure 9. Ensemble averaged gust spectra for wind speeds at (a) 13 ft (4m) and (b) 33 ft (10m) heights above ground. Empirical spectra (dashed lines) are given by $4St_u(1 + 70.8 St_u^2)^{-5/6}$ where the Strouhal Number $St_u = fL_x/\bar{V}$.

access to a field experiment. In the past two and a half years, since the laboratory became fully operational, three distinguished researchers in wind engineering have spent 4-6 months at Texas Tech University. These visits have allowed investigators the opportunity to pursue their own interests as related to full scale testing, as well as provide valuable input toward the continuing improvement of the Wind Engineering Research Field Laboratory.

7. ACKNOWLEDGEMENTS

The significant contributions of the advisory committee members are greatly appreciated. Past members of the advisory committee include Richard Marshall, Henry Liu, Tim Reinhold, Dale Perry, Dick Kennon, and Jon Peterka. Dave Surry, Robert Scanlan and Joe Minor, current members of the advisory committee, have contributed significantly to the progress made at the field laboratory. Researchers such as Jack Cermak, John Holmes, Hisashi Okada and Chris Letchford have worked at the site and contributed directly in improving the field laboratory. The donation of the test building by Butler Manufacturing Co. and the undercarriage and rail by American Buildings Co. is appreciated. Financial support from the U.S. National Science Foundation and the Texas Advanced Research Program is acknowledged .

8. REFERENCES

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